The Solutions 2024

B—bookwork, A—application, E—new example, T—new theory

1.

a) Distribution: $p(X=0)=\frac{2}{3}$ $p(X=1)=\frac{1}{3}$

$p(Y=0)=\frac{1}{3}$ $p(Y=1)=\frac{2}{3}$

i) $H(X)=\frac{2}{3}\log\frac{3}{2}+\frac{1}{3}\log 3 = 0.918$ bits $= H(Y)$ [3B]

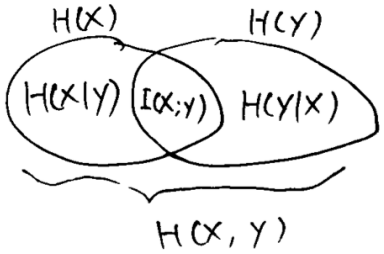
ii) $H(X|Y)=\frac{1}{3}H(X|Y=0)+\frac{2}{3}H(X|Y=1)$ [3B]

$=\frac{1}{3}\times 0+\frac{2}{3}\times 1=\frac{2}{3}=0.667$ bits $= H(Y|X)$

iii) $H(X,Y)=3\times\frac{1}{3}\log 3=\log 3=1.585$ bits [3B]

iv) $I(X;Y)=H(X)-H(X|Y)=0.918-0.667=0.251$ bits [3B]

v)



[3B]

b) $X = n$ means that Tail occurs for the first $n-1$ flips, while Head occurs for the n-th flip. Thus

$$P(X = n) = \left(\frac{1}{2}\right)^{n-1} \frac{1}{2} = \left(\frac{1}{2}\right)^{n} \qquad \text{[5 E]}$$

i) $H(X) = \sum_{n=1}^{\infty} 2^{-n} \log 2^{n} = \sum_{n=1}^{\infty} n \cdot 2^{-n} \cdot \log 2 = \frac{\frac{1}{2}}{(1-\frac{1}{2})^2} = 2$ bits

ii) Ask if $X = 1, 2, 3, \cdots$ in turn, i.e., [5E]

Is $X = 1$?

If not, is $X = 2$?

If not, is $X = 3$?

...

Expected number of questions $= \sum_{n=1}^{\infty} n \left(\frac{1}{2}\right)^{n} = 2$

2.

a) i)

(1) chain rule [1B]

(2) chain rule in another way [1B]

(3) $H(e|x,y) \geq 0$ entropy is non-negative

$H(e|y) \leq H(e)$ conditioning reduces entropy [2B]

(4) total probability theorem [1B]

(5) $H(e) = H(p_e)$

Given y and $e=0$, $X=y$, so entropy $= 0$.

Given y and $e=1$, $X \neq y$ but can take any of the $|X|-1$ values, so entropy $\leq \log(|X|-1)$ [3B]

(6) algebra [1B]

(7) $H(p_e) \leq 1$ [1B]

b) H(X) = H(p) = H(0.11) = 0.5

i) U_1 is a Bernoulli random variable with parameter [2A]

$$p_1 = 2p(1-p) = 0.196$$

So

$$H(U_1) = H(p_1) = 0.713$$

[2A]

$$H(U_2|U_1) = 2H(X) - H(U_1) = 1 - 0.713 = 0.287$$

[2A]

ii)

Now, what is the conditional distribution $P(U_2|U_1)$? Using Bayes' rule, we have [2T]

$$P(U_2|U_1) = \frac{P(U_2,U_1)}{P(U_1)} = \frac{P(U_1|U_2)P(U_2)}{P(U_1)} = \frac{P(U_1|X_2)P(X_2)}{P(U_1)}$$

We have shown above that U_1 is $Bernoulli(p_1)$. Hence [4T]

$$P(U_2 = 0|U_1 = 0) = \frac{P(X_1 = 0)P(X_2 = 0)}{P(U_1 = 0)} = \frac{(1-p)^2}{1-p_1} = 0.985$$

$$P(U_2 = 1|U_1 = 0) = \frac{P(X_1 = 1)P(X_2 = 1)}{P(U_1 = 0)} = \frac{p^2}{1 - p_1} = 0.015$$

$$P(U_2 = 0|U_1 = 1) = \frac{P(X_1 = 1)P(X_2 = 0)}{P(U_1 = 1)} = \frac{p(1-p)}{p_1} = 0.5$$

$$P(U_2 = 1|U_1 = 1) = \frac{P(X_1 = 0)P(X_2 = 1)}{P(U_1 = 1)} = \frac{p(1-p)}{p_1} = 0.5$$

Now we can also compute the conditional entropy $H(U_2|U_1)$, as average row entropy:

$$\begin{aligned} H(U_2|U_1) &= H(U_2|U_1 = 0)P(U_1 = 0) + H(U_2|U_1 = 1)P(U_1 = 1) \\ &= H(U_2|U_1 = 0) \times 0.804 + H(U_2|U_1 = 1) \times 0.196 \\ &= H(0.015) \times 0.804 + 1 \times 0.196 = 0.112 \times 0.804 + 0.196 \\ &= 0.09 + 0.196 = 0.286 \end{aligned}$$

which agrees with the result in (i). [3T]

3.

a)

(1) $X_i = f_i(W, Y_{1:i-1})$, $Z_j = Y_j - X_j$ $1 \le j \le i-1$ [1B]

(2) $Y_i = X_i + Z_i$ Shift doesn't change entropy [1B]

(3) given $Z_{1:i-1}$, Z_i and $(W, Y_{1:i-1}, X_{1:i})$ are conditionally independent. [2B]

(4) $h(Y) \le \frac{1}{2}\log|2\pi e K_Y|$ Gaussian has maximum entropy. [2B]

$h(Z) = \frac{1}{2}\log|2\pi e K_Z|$ [1B]

(5) definition [1B]

(6) $|2K| = 2^n |K|$ for matrix K [1B]

(7) algebra [1B]

(8) $C_n = \max \frac{1}{2n} \log \frac{|K_X + K_Z|}{|K_Z|}$ [1B]

b) Since the channel is symmetric, it is easy to compute its capacity: [3E]

$H(Y|X) = 1$
$I(X;Y) = H(Y) - H(Y|X) = H(Y) - 1.$

So mutual information is maximized when Y is uniformly distributed, which occurs when the input X is uniformly distributed. Therefore the capacity in bits is [2E]

$C = \log_2 5 - 1 = \log_2 2.5 = 1.32.$

c) Let the probabilities of the three input symbols be p_1, p_2 and p_3. Then the probabilities of the three output symbols can be easily calculated to be $(2/3p_1 + 1/3p_2, 1/3, 1/3p_2 + 2/3p_3)$, and therefore [3T]

$$I(X;Y) = H(Y) - H(Y|X) = H\left(\frac{2}{3}p_1 + \frac{1}{3}p_2, \frac{1}{3}, \frac{1}{3}p_2 + \frac{2}{3}p_3\right) - (p_1 + p_3)H\left(\frac{2}{3}, \frac{1}{3}\right) - p_2 \log 3$$
$$= H(\frac{1}{3} + \frac{1}{3}(p_1 - p_3), \frac{1}{3}, \frac{1}{3} - \frac{1}{3}(p_1 - p_3)) - (p_1 + p_3)H(\frac{2}{3}, \frac{1}{3}) - (1 - p_1 - p_3)\log 3$$

where we have substituted $p_2 = 1 - p_1 - p_3$. Now if we fix $p_1 + p_3$, then the second and third terms are fixed, and the first term is maximized if $p_1 - p_3 = 0$, i.e., if $p_1 = p_3$.

Now setting $p_1 = p_3$, we have [3T]

$$I(X;Y) = H\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) - (p_1 + p_3)H\left(\frac{2}{3}, \frac{1}{3}\right) - (1 - p_1 - p_3)\log 3$$
$$= \log 3 - (p_1 + p_3)H\left(\frac{2}{3}, \frac{1}{3}\right) - (1 - p_1 - p_3)\log 3 = (p_1 + p_3)(\log 3 - H\left(\frac{2}{3}, \frac{1}{3}\right))$$

which is maximized if $p_1 + p_3$ is as large as possible.

Therefore the maximizing distribution corresponds to $p_1 + p_3 = 1$, $p_1 = p_3$, and therefore $(p_1, p_2, p_3) = (1/2, 0, 1/2)$. [2T]

The capacity of this channel [2T]

$$C = \log 3 - H\left(\frac{2}{3}, \frac{1}{3}\right) = \log 3 - \left(\log 3 - \frac{2}{3}\right) = \frac{2}{3}$$

4.

a)

i) Multi-access channel is a many-to-one channel where there are many senders but only one receiver. A typical example is the uplink of a cellular communication system. [3B]



[2B]

ii)

$$R_1 \le W \log\left(1+\frac{P_1}{N\,W}\right)$$

$$R_2 \le W \log\left(1+\frac{P_2}{N\,W}\right) \quad [3B]$$

$$R_1+R_2 \le W \log\left(1+\frac{P_1+P_2}{N\,W}\right)$$

The capacity region is a pentagon. To achieve corner points, one may use onion peeling. [2B]

iii) If W goes to infinity, then

$$R_1 \to \frac{P_1}{N}$$

$$R_2 \to \frac{P_2}{N} \quad [3T]$$

$$R_1+R_2 \to \frac{P_1}{N}+\frac{P_2}{N}$$

where the unit is nat (there is an extra factor log(e) if the unit is bit). Therefore, senders can transmit as if there were no interference. [2T]

b) The Slepian-Wolf region is given by [2B]

$$R_1 \ge H(X\,|\,Y)$$

$$R_2 \ge H(Y\,|\,X)$$

$$R_1+R_2 \ge H(X,Y)$$

X = Bernoulli(p). $Y = X \oplus Z$, Z = Bernoulli(r).

Then Y = Bernoulli($p*r$), where $p*r = p(1-r)+r(1-p)$. [2E]

$H(X) = H(p)$.

$H(Y) = H(p*r)$,

$H(X,Y) = H(X,Z) = H(X)+H(Z) = H(p)+H(r)$. [2E]

Hence $H(Y|X) = H(r)$ and $H(X|Y) = H(p)+H(r)- H(p*r)$. [2E]

This region is shown below. [2E]

